Hematology RBC / Anemia / Hemolysis Codex Guide

- Block: Hematology
- Method: Codex-authored manual distillation from extracted lecture markdown in this workspace.
- Sources:
  - 20210303_Moliterno_B12_FolateMetabolismLectureMoliterno.pdf
  - 20210303_Naik_RBC_Enzymatic_Pathways_Congenital_Enzymopathies.pdf
  - 20210303_Naik_RBC_Membrane_Structure_Congenital_Membranopathies.pdf
  - 20210303_Brodsky_Paroxysmal_Nocturnal_Hemoglobinuria.pdf
  - 20220307_Naik_Pathophysiology_Sickle_Cell_Disease.pdf
  - 2025_Naik_Clinical_Complications_Sickle_Cell_Disease.pdf
  - 20230307_Naik_Heme_Synthesis_Sideroblastic_Anemias.pdf
  - 20230307_Naik_Hepatic_Porphyrias.pdf

One-Minute Mental Model

RBC disease questions usually become manageable if you first decide which part of the red cell system is broken:

1. **DNA synthesis problem** -> macrocytosis/megaloblastosis: B12 or folate deficiency.
2. **RBC intrinsic structural problem** -> congenital hemolysis: membrane, enzyme, or hemoglobin defect.
3. **Complement-protection problem** -> acquired intravascular hemolysis: PNH.
4. **Hemoglobin production problem** -> microcytosis/hypochromia: iron deficiency, thalassemia, anemia of chronic disease, sideroblastic anemia, or lead poisoning.
5. **Heme precursor handling problem** -> porphyrias: hepatic symptoms without anemia, or erythroid heme synthesis failure with sideroblastic anemia.

The most testable fork is whether anemia is **hypoproliferative** or **hyperproliferative**. Hemolysis usually has reticulocytosis because the marrow is trying to compensate. Ineffective erythropoiesis and production disorders often have an inappropriately low or normal reticulocyte response.

Highest-Yield Concepts

B12 and Folate Deficiency

- B12 deficiency classically causes **megaloblastic anemia, gastric atrophy, and neurologic degeneration**. [B12/Folate, slide 3]
- Macrocytic/megaloblastic anemia happens because slowed DNA division makes rapidly dividing marrow cells large and ineffective; marrow may be hypercellular while peripheral production is poor. [B12/Folate, slides 6-8]
- B12 absorption depends on intrinsic factor and the terminal ileum. Common B12 deficiency mechanisms are malabsorption from loss of gastric function, intrinsic factor deficiency, atrophic gastritis, gastrectomy, aging, H2 blockers, pancreatic disease, and terminal ileal disease. [B12/Folate, slides 11-12]
- Dietary B12 deficiency is less common but occurs in vegans and breast-fed infants of vegan or B12-deficient mothers. [B12/Folate, slide 13]
- B12 deficiency can present as anemia with neurologic symptoms, anemia without neurologic symptoms, or neurologic symptoms with little/no anemia. [B12/Folate, slide 16]
- Low serum B12 plus neurologic disease and elevated methylmalonic acid supports B12 deficiency. [B12/Folate, slide 14]
- Folate deficiency develops faster because body stores are smaller; folate is absorbed in the proximal jejunum and is found in leafy greens, asparagus, broccoli, spinach, liver, kidney, fruits, and mushrooms. [B12/Folate, slide 19]
- Folate deficiency causes include poor intake, malabsorption, drugs such as dilantin/barbiturates, DHFR inhibitors, B12 loss, pregnancy, infancy, and hemolytic anemia. [B12/Folate, slide 22]
- The methyl-folate trap matters clinically: folate can improve anemia in B12 deficiency while neurologic damage continues, so **rule out B12 deficiency before treating folate deficiency alone**. [B12/Folate, slides 23 and 28]
- Folate deficiency is associated with fetal neural tube defects; grain fortification was designed to raise daily folic acid intake. [B12/Folate, slide 25]

Congenital Hemolytic Anemia Framework

- Congenital hemolytic anemias fall into three intrinsic RBC defect groups: **membranopathies, enzymopathies, and hemoglobinopathies**. [RBC Enzymes, slide 6; RBC Membrane, slide 8; Sickle Pathophysiology, slide 6]
- Hemolysis can be classified by congenital vs acquired, intravascular vs extravascular, and intrinsic vs extrinsic RBC defect. [RBC Membrane, slide 6]
- Mature RBCs have no nucleus or mitochondria. This creates room for hemoglobin and improves deformability, but it forces RBCs to rely on glycolysis for ATP and shunt pathways for specialized functions. [RBC Enzymes, slides 7-9]

G6PD Deficiency

- G6PD deficiency is X-linked and causes reduced NADPH, reduced glutathione, and vulnerability to oxidative injury. [RBC Enzymes, slide 13]
- RBCs are especially affected because they lack nuclei/mitochondria and depend on the hexose monophosphate shunt for oxidative protection. [RBC Enzymes, slide 14]
- Oxidative injury denatures hemoglobin into Heinz bodies; splenic macrophages remove Heinz bodies, producing **bite cells and blister cells**. [RBC Enzymes, slides 15-16]
- G6PD hemolysis is episodic, usually only during oxidant stress. Common triggers include antimalarials, sulfa drugs, acute infection/sepsis, and fava beans. [RBC Enzymes, slide 18]
- Treatment is removal/avoidance of the offending agent; severe anemia may require transfusion. [RBC Enzymes, slide 18]
- G6PD activity testing can be falsely normal during acute hemolysis because older, enzyme-poor RBCs have already lysed and reticulocytes have higher G6PD activity. Test again after recovery if suspicion remains high. [RBC Enzymes, slide 19]

Pyruvate Kinase Deficiency

- Pyruvate kinase deficiency is autosomal recessive and affects erythrocyte glycolysis, causing decreased ATP and increased upstream metabolites. [RBC Enzymes, slide 20]
- Increased upstream flow into the 2,3-DPG shunt raises 2,3-DPG, right-shifting the oxygen dissociation curve and allowing patients to tolerate more severe anemia than expected. [RBC Enzymes, slide 21]
- Low ATP damages RBC structure/shape, often producing acanthocytes, and causes extravascular hemolysis. [RBC Enzymes, slide 22]
- PK deficiency can cause moderate/severe hemolytic anemia from birth, neonatal hyperbilirubinemia, kernicterus risk, pigmented gallstones, splenomegaly, and aplastic crisis with parvovirus. [RBC Enzymes, slides 23-25]
- Treatments include phototherapy for neonatal hyperbilirubinemia, transfusions for severe symptomatic anemia, erythropoietin to augment reticulocytosis, and splenectomy to reduce hemolysis. [RBC Enzymes, slides 24-25]

Hereditary Spherocytosis and Membranopathies

- The RBC membrane includes a lipid bilayer, cell surface proteins, transmembrane proteins, carbohydrates, and a submembrane cytoskeleton. The cytoskeleton maintains biconcave shape and deformability. [RBC Membrane, slide 10]
- Congenital membranopathies are caused by deficiency of cytoskeletal proteins or proteins that connect cytoskeleton to lipid bilayer, especially band proteins, ankyrin, and spectrin. [RBC Membrane, slides 14-16]
- Hereditary spherocytosis is usually autosomal dominant and can involve ankyrin-1, beta-spectrin, band 3, alpha-spectrin, or protein 4.2 defects. [RBC Membrane, slide 16]
- Loss of vertical interactions between lipid bilayer and cytoskeleton produces loss of biconcave shape and extravascular hemolysis. [RBC Membrane, slide 17]
- Spherocytes have loss of central pallor, increased MCHC, reduced deformability, reduced stability, and reduced surface-area-to-volume ratio. [RBC Membrane, slide 18]
- Osmotic fragility is increased in spherocytic disorders because the cells are less stable under osmotic stress. [RBC Membrane, slide 19]
- Chronic hereditary spherocytosis causes mild/moderate hemolysis since birth, pigmented gallstones, splenomegaly, and parvovirus-associated aplastic crisis. [RBC Membrane, slide 20]
- Severe uncompensated hereditary spherocytosis can be treated with transfusions, erythropoietin, or splenectomy; vaccinate before splenectomy because of risk from encapsulated organisms. [RBC Membrane, slide 21]
- HS and warm AIHA both show spherocytes and extravascular hemolysis, but HS is congenital/intrinsic/DAT negative, while warm AIHA is acquired/extrinsic/DAT positive. [RBC Membrane, slide 22]

PNH

- PNH is an acquired PIGA mutation in a hematopoietic stem cell; PIGA is needed for GPI-anchor biosynthesis, so hematopoietic cells lose GPI-anchored proteins. [PNH, slide 5]
- CD55 blocks C3 convertase and CD59 prevents MAC formation. Loss of CD55/CD59 leaves cells vulnerable to complement-mediated destruction. [PNH, slide 7]
- PNH causes rapid **intravascular hemolysis** with very high LDH, undetectable haptoglobin, hemoglobinuria, and a negative DAT. The smear may lack classic findings because cells are destroyed quickly. [PNH, slides 3-4 and 9]
- Diagnosis is flow cytometry showing CD55/CD59-negative cells in both RBCs and WBCs, because the mutation is in the hematopoietic stem cell. [PNH, slide 10]
- Free hemoglobin depletes nitric oxide, impairing smooth muscle relaxation and vasodilation. Consequences include dysphagia, abdominal cramping, erectile dysfunction, renal pigment injury, and thrombosis. [PNH, slide 11]
- Eculizumab is an anti-C5 monoclonal antibody that prevents C5 cleavage and MAC formation, decreasing hemolysis, symptoms, and thrombosis risk. It increases Neisseria infection risk, so vaccinate before therapy. [PNH, slide 12]

Sickle Cell Disease

- HbS is caused by a beta-globin point mutation on chromosome 11, substituting valine for glutamic acid. HbS is a hemoglobin variant with different molecular properties. [Sickle Pathophysiology, slide 9]
- HbS polymerizes when deoxygenated; beta-S chains bind adjacent beta-S chains, forming long rods, elongating RBCs, increasing rigidity, and decreasing stability. [Sickle Pathophysiology, slides 15-16]
- Sickling causes both extravascular hemolysis, intravascular hemolysis, and vaso-occlusion in small/medium arteries. [Sickle Pathophysiology, slide 17; Sickle Complications, slide 10]
- Polymerization depends on percent HbS: no polymerization with HbS below 50%, so HbAS trait generally does not sickle clinically; HbSS has >90% HbS and no HbA. [Sickle Pathophysiology, slide 18]
- Sickle cell disease includes any genotype that causes HbS polymerization: HbSS, HbSC, HbS beta-plus thalassemia, and HbS beta-zero thalassemia. [Sickle Pathophysiology, slide 19; Sickle Complications, slide 9]
- Sickle trait is HbAS with about 60% HbA and 40% HbS, no significant symptoms, normal CBC, and malaria protection. [Sickle Pathophysiology, slides 11-12]
- HbSC has about 50% HbS and 50% HbC, no HbA; HbC causes intracellular dehydration that promotes HbS polymerization, but sickling is usually less severe than HbSS. [Sickle Pathophysiology, slides 20-22]
- S beta-plus thalassemia has HbS >60% and HbA <40%, with increased HbA2/HbF; S beta-zero thalassemia has HbS >90%, no HbA, and resembles HbSS. [Sickle Pathophysiology, slides 23-24]
- Sickle solubility screening detects presence of HbS but cannot distinguish trait from disease; electrophoresis and HPLC differentiate/quantify hemoglobin patterns. [Sickle Pathophysiology, slides 25-27]
- Symptoms are absent at birth because HbF persists; disease appears in mid/late infancy as HbS rises after the hemoglobin switch. [Sickle Complications, slide 8]
- Hemolysis severity is genotype-dependent: HbSS/S beta-zero are most severe; HbSC/S beta-plus are less severe. [Sickle Complications, slide 11]
- Vaso-occlusive stressors include hypoxia, infection, and dehydration. Acute complications include pain crises, stroke, acute chest syndrome, and splenic sequestration. [Sickle Complications, slides 12-15]
- Acute chest syndrome creates a vicious cycle: lung vaso-occlusion causes hypoxia, hypoxia increases deoxygenated HbS, deoxygenated HbS polymerizes, and sickling worsens lung vaso-occlusion. [Sickle Complications, slide 14]
- Chronic splenic autoinfarction is common, often by early childhood, and increases risk of encapsulated organisms. Howell-Jolly bodies mark splenic dysfunction. Children receive penicillin prophylaxis to reduce Streptococcus pneumoniae risk. [Sickle Complications, slide 16]
- Severe acute complications can be treated with hydration, oxygen, and exchange transfusion, which removes HbS-containing cells and replaces them with HbAA donor RBCs. [Sickle Complications, slide 17]
- Hydroxyurea increases HbF above 15-20%, lowering relative HbS, reducing polymerization/sickling, improving anemia, and reducing complications. [Sickle Complications, slide 18]

Sideroblastic Anemia and Lead Poisoning

- Microcytic hypochromic anemia reflects decreased hemoglobin production. Causes include thalassemias, iron deficiency, anemia of chronic disease, lead poisoning, and sideroblastic anemia. [Heme Synthesis, slide 6]
- Erythroid heme synthesis begins and ends in mitochondria, occurs in erythroblasts/nucleated RBCs, and cannot occur in mature RBCs because they lack mitochondria. [Heme Synthesis, slide 7]
- The first erythroid heme synthesis step uses ALAS-2 to combine succinyl-CoA and glycine into ALA; the last

important step incorporates iron into protoporphyrin to form heme. [Heme Synthesis, slide 8]

- Sideroblastic anemias are caused by congenital or acquired defects in heme synthesis enzymes/cofactors, causing decreased heme production and mitochondrial iron accumulation in erythroblasts: **ring sideroblasts**. [Heme Synthesis, slide 9]
- Anemia comes from low hemoglobin production plus ineffective erythropoiesis because iron toxicity causes erythroblast apoptosis in marrow. [Heme Synthesis, slide 10]
- X-linked sideroblastic anemia is the most common congenital form; it is X-linked recessive, caused by ALAS-2 mutation, and usually causes mild-moderate deficiency rather than total absence. [Heme Synthesis, slide 11]
- Diagnosis requires marrow biopsy showing ring sideroblasts; genetic testing can confirm the mutation. It can mimic iron deficiency or thalassemia on smear. [Heme Synthesis, slide 12]
- Vitamin B6 is a cofactor for ALAS-2, so pyridoxine improves anemia in many patients; severe symptomatic anemia may require intermittent transfusions. [Heme Synthesis, slide 13]
- Chronic sideroblastic anemia can cause iron overload from ineffective erythropoiesis and transfusions; ineffective erythropoiesis lowers hepcidin, raises ferroportin, and increases iron absorption. Treatment is iron chelation. [Heme Synthesis, slide 14]
- Acquired sideroblastic anemia causes include lead poisoning, heavy alcohol use, isoniazid, and B6 deficiency. [Heme Synthesis, slide 15]
- Lead suppresses enzymes converting ALA to PBG and protoporphyrin to heme, causing precursor accumulation. Testing includes serum lead and serum/urine ALA or protoporphyrin. [Heme Synthesis, slide 16]
- Lead poisoning causes sideroblastic anemia and basophilic stippling; stippling is aggregated ribosomal RNA, not iron. [Heme Synthesis, slide 18]
- Lead also causes neurotoxicity: in-utero neurodevelopmental delay, childhood seizures/permanent neuropsychiatric changes, and adult memory/personality changes. [Heme Synthesis, slide 19]
- Treatment is source identification and exposure reduction for chronic toxicity; acute toxicity may require chelation. [Heme Synthesis, slide 20]

Hepatic Porphyrias

- Liver heme synthesis uses ALAS-1, while erythroid heme synthesis uses ALAS-2. Hepatic heme is used for cytochrome P450 drug metabolism. [Porphyrias, slide 5]
- Hepatic porphyrias are enzyme defects in liver heme synthesis; symptoms come from toxic upstream precursor accumulation, not anemia. [Porphyrias, slides 6 and 15]
- ALAS-1 is induced by progesterone, alcohol, rifampin, antiepileptics, and other cytochrome P450 inducers. Induction worsens precursor accumulation and makes symptoms episodic. [Porphyrias, slide 7]
- AIP is the most common congenital hepatic porphyria, autosomal dominant with incomplete penetrance, and involves impaired conversion of PBG to the next metabolite. ALA and PBG accumulate during attacks and are neurotoxic. [Porphyrias, slide 8]
- AIP presents with intermittent non-localized abdominal pain, extremity numbness/weakness, and neuropsychiatric symptoms. Diagnosis requires suspicion; spot urine ALA and PBG are high during attacks and may normalize between attacks. [Porphyrias, slide 9]
- AIP treatment includes avoiding ALAS-1 inducers and giving IV hemin during acute attacks. Hemin enters hepatocytes and shuts off ALAS-1 by feedback, lowering ALA/PBG production. [Porphyrias, slide 10]
- Hemin is for acute attacks because repeated use risks renal toxicity and iron overload; givosiran inhibits ALAS-1 for prophylaxis in frequent attacks, but hemin is still used for breakthrough attacks. [Porphyrias, slide 11]
- PCT is the most common acquired hepatic porphyria, associated with hepatic iron overload and alcohol use, and causes chronic photosensitivity with blistering skin lesions plus liver disease. [Porphyrias, slides 12-13]
- PCT diagnosis uses high total porphyrins in plasma or urine. Treatment includes avoiding ALAS-1 inducers, specifically avoiding alcohol, and phlebotomy to reduce ferritin to low-normal range. [Porphyrias, slides 13-14]

Must-Know Comparisons

Disorder	Mechanism	Key clues	Test	Treatment anchor
B12 deficiency	Impaired DNA synthesis; often malabsorption	Macrocytosis, pancytopenia, neuropathy, loss of vibration/position sense	Low B12; methylmalonic acid supports	Parenteral or high-dose oral B12; identify malabsorption

Disorder	Mechanism	Key clues	Test	Treatment anchor
Folate deficiency	Impaired DNA synthesis	Macrocytosis without B12 neuro findings; pregnancy/poor diet/drugs	Folate assessment; rule out B12	Oral folate; diet; treat bowel disease
G6PD deficiency	Low NADPH/glutathione under oxidant stress	Episodic hemolysis after sulfa/antimalarial/fava/infection; bite/blister cells	G6PD activity, repeated after episode if needed	Avoid trigger; transfuse if severe
PK deficiency	Low ATP from glycolysis defect	Chronic hemolysis from birth, neonatal jaundice, acanthocytes, high 2,3-DPG	Enzyme/genetic testing	Phototherapy, transfusion, EPO, splenectomy if severe
Hereditary spherocytosis	Cytoskeleton/membrane tethering defect	Spherocytes, high MCHC, congenital hemolysis, splenomegaly	Increased osmotic fragility; DAT negative	Supportive; splenectomy if severe
Warm AIHA	Autoantibody-mediated extrinsic hemolysis	Spherocytes but acquired	DAT positive	Treat immune cause
PNH	Acquired PIGA mutation causing loss of CD55/CD59	Intravascular hemolysis, hemoglobinuria, thrombosis, dysphagia/abdominal pain	Flow cytometry for CD55/CD59-negative cells	Eculizumab; Neisseria vaccination
HbSS	HbS >90%, no HbA	Severe sickling, hemolysis, vaso-occlusion	HPLC/electrophoresis	Hydroxyurea; exchange transfusion for severe acute events
HbAS trait	HbA ~60%, HbS ~40%	Usually asymptomatic, normal CBC	HPLC/electrophoresis	Usually none
Sideroblastic anemia	Failed heme synthesis with mitochondrial iron	Microcytic anemia with normal/high iron; ring sideroblasts	Bone marrow Prussian blue	B6 if ALAS-2; transfusion/chelation as needed
Lead poisoning	Blocks ALA -> PBG and protoporphyrin -> heme	Basophilic stippling, neurotoxicity	Lead level; ALA/protoporphyrin	Remove exposure; chelation if acute
AIP	Hepatic precursor buildup: ALA/PBG	Episodic abdominal pain, weakness, neuropsychiatric symptoms, normal CBC	Spot urine ALA/PBG during attack	Avoid triggers; IV hemin
PCT	Acquired hepatic porphyrin buildup	Chronic blistering photosensitivity, liver disease	High plasma/urine total porphyrins	Avoid alcohol/triggers; phlebotomy

Common Confusions To Avoid

- **Macrocytosis is not always "just folate."** B12 deficiency can be neurologically devastating; rule it out before folate-only treatment.
- **G6PD testing during the crisis can mislead you.** A normal level during acute hemolysis does not fully exclude it.
- **G6PD is episodic; PK deficiency is chronic.** G6PD needs oxidative stress; PK deficiency causes ongoing energy failure.
- **HS and warm AIHA both make spherocytes.** DAT separates them: HS is DAT negative, warm AIHA is DAT positive.
- **PNH is acquired despite involving an X-linked gene.** The mutation occurs in hematopoietic stem cells, so both males and females can be affected.
- **Sickle solubility screen detects HbS, not disease severity.** Use electrophoresis/HPLC to distinguish trait vs disease genotypes.
- **HbSC can sickle even with only about 50% HbS.** HbC dehydrates the RBC and promotes polymerization.
- **Sideroblastic anemia is not iron deficiency.** Iron is present but trapped in erythroblast mitochondria.
- **Hepatic porphyrias are not anemias.** The liver pathway uses different enzymes from erythroid heme synthesis.
- **AIP attacks can have normal labs/imaging unless you think of it.** Spot urine ALA/PBG during attack is the key clue.

Rapid Review

- B12 deficiency: macrocytic megaloblastic anemia + neurologic degeneration; malabsorption is the common mechanism; methylmalonic acid helps confirm.

- Folate deficiency: macrocytic anemia, neural tube defect risk, no classic B12 neuro syndrome; rule out B12 before folate-only replacement.
- G6PD: X-linked, oxidative stress, Heinz bodies -> bite/blister cells, false-normal test during acute hemolysis.
- PK deficiency: AR, low ATP, acanthocytes, extravascular hemolysis, high 2,3-DPG helps tolerate anemia.
- HS: congenital membrane tethering defect, spherocytes, high MCHC, increased osmotic fragility, DAT negative.
- PNH: PIGA mutation, absent GPI anchors, loss of CD55/CD59, intravascular hemolysis, thrombosis, flow diagnosis, eculizumab.
- Sickle disease: deoxygenated HbS polymerization causes hemolysis and vaso-occlusion; HPLC/electrophoresis define genotype.
- Acute sickle complications: pain crisis, stroke, acute chest syndrome, splenic sequestration.
- Chronic sickle complication: splenic autoinfarction with encapsulated organism risk and Howell-Jolly bodies.
- Sideroblastic anemia: heme synthesis failure, ring sideroblasts, microcytosis with iron trapped in mitochondria.
- Lead: sideroblastic anemia + basophilic stippling + neurotoxicity.
- AIP: episodic abdominal pain/weakness/neuropsychiatric symptoms, high urine ALA/PBG during attack, hemin.
- PCT: chronic blistering photosensitivity/liver disease, high total porphyrins, phlebotomy and alcohol avoidance.

Active Recall Questions

1. Why can B12 deficiency produce both anemia and neurologic findings?
2. Why is folate replacement dangerous if B12 deficiency has not been excluded?
3. What makes mature RBCs uniquely vulnerable to G6PD deficiency?
4. Why can G6PD activity be falsely normal during an acute hemolytic episode?
5. How does PK deficiency increase 2,3-DPG, and why does that change symptoms?
6. What membrane/cytoskeleton defects cause hereditary spherocytosis?
7. How do you separate hereditary spherocytosis from warm autoimmune hemolytic anemia?
8. Why does PNH cause DAT-negative intravascular hemolysis?
9. How does free hemoglobin create dysphagia, abdominal pain, erectile dysfunction, and thrombosis in PNH?
10. Why does HbAS trait usually not produce clinically significant sickling?
11. Compare HbSS, HbSC, HbS beta-plus thalassemia, and HbS beta-zero thalassemia by HbS/HbA pattern.
12. Why do sickle cell symptoms begin in infancy rather than at birth?
13. Explain the acute chest syndrome vicious cycle.
14. Why does hydroxyurea reduce sickle cell complications?
15. What distinguishes sideroblastic anemia from iron deficiency anemia?
16. Why does lead poisoning cause both anemia and neurologic symptoms?
17. Why do hepatic porphyrias not cause anemia?
18. What triggers AIP attacks, and why does hemin work?
19. How are AIP and PCT different in symptoms, diagnosis, and treatment?
20. If you see microcytosis with normal iron studies and normal hemoglobin electrophoresis, what diagnoses should move up your list?